

2048 Dark Edition

Setup Guide — What YOU Need to Do

So Claude Code can build & deliver your APK automatically.

What This Guide Is

Claude Code is an AI agent that runs inside your terminal and writes code, runs commands, and builds projects on its own. But it can only work with what is already installed on your computer. This guide tells you exactly what to install once, so Claude Code can take over and build the entire 2048 APK without you needing to code anything.

You only do this setup **ONCE**. After that, Claude Code does everything.

Step 1 — Install Android Studio

Android Studio is the only big download you need. It bundles the Android SDK, the emulator, Java JDK 17, and all build tools in one installer.

1

Download Android Studio

- Go to: <https://developer.android.com/studio>
- Click the big green Download button
- Choose the version for your OS: Windows / macOS / Linux

2

Install It

- Windows: run the .exe installer, click Next through everything
- macOS: drag Android Studio into Applications folder
- Linux: unzip and run studio.sh from the bin/ folder
- On first launch it will run a Setup Wizard — complete it

3

Complete the Setup Wizard (important!)

- Wizard opens automatically on first launch
- Select: Standard installation (not Custom)
- It will download: Android SDK, Build Tools, Emulator
- This download is ~1.5 GB — keep Android Studio open until done
- When it says Finish — you're done with this step

Step 2 — Install Required SDK Components

Inside Android Studio, open the SDK Manager and make sure these are installed:

Component	Where to find it	Status needed
Android 14 (API 34)	SDK Platforms tab	✓ Installed
Android 5.0 (API 21)	SDK Platforms tab	✓ Installed
Android SDK Build-Tools 34.x	SDK Tools tab	✓ Installed

Android SDK Platform-Tools	SDK Tools tab	✓ Installed
Android Emulator	SDK Tools tab	Optional

How to open SDK Manager:

Android Studio → top menu → Tools → SDK Manager (or the cube icon in toolbar)

Step 3 — Verify Java JDK 17

Java JDK 17 comes bundled with Android Studio. You do NOT need to install it separately. But you need to make sure your terminal can find it.

1 Open your terminal (Command Prompt / Terminal / PowerShell)

- Windows: press Win+R, type cmd, press Enter
- macOS: press Cmd+Space, type Terminal, press Enter
- Linux: open your terminal application

2 Run this command to check Java:

- Type exactly: `java -version`
- Press Enter
- You should see something like: `openjdk 17.0.x ...`
- If it says 'command not found' — see note below

```
$ java -version
openjdk version "17.0.10" 2024-01-16
OpenJDK Runtime Environment (build 17.0.10+7)
```

If java -version says 'not found':

Android Studio bundles JDK 17 at this path:

```
Windows: C:\Program Files\Android\Android Studio\jbr\
macOS:   /Applications/Android Studio.app/Contents/jbr/Contents/Home/
Linux:   ~/android-studio/jbr/
```

Add the bin/ folder inside that path to your system PATH variable. Search online for 'how to add to PATH on Windows/macOS/Linux' if unsure.

Step 4 — Set ANDROID_HOME Environment Variable

Claude Code needs to know where your Android SDK is installed. You tell it by setting one environment variable called `ANDROID_HOME`.

1 Find your SDK path in Android Studio

- Open Android Studio → Tools → SDK Manager
- At the top it shows: Android SDK Location
- Copy that full path (e.g. `C:\Users\You\AppData\Local\Android\Sdk`)

2 Set ANDROID_HOME — Windows

- Search: Edit the system environment variables
- Click: Environment Variables
- Under System variables click New
- Variable name: ANDROID_HOME
- Variable value: paste your SDK path
- Click OK on all dialogs
- Also add to PATH: %ANDROID_HOME%\platform-tools

3 Set ANDROID_HOME — macOS / Linux

- Open terminal and run:
- `echo 'export ANDROID_HOME=$HOME/Library/Android/sdk' >> ~/.zshrc`
- `echo 'export PATH=$PATH:$ANDROID_HOME/platform-tools' >> ~/.zshrc`
- `source ~/.zshrc`
- (Use `~/.bashrc` instead if you use bash, not zsh)

4 Verify it worked

- Open a NEW terminal window (important — old ones won't have the new variable)
- Run: `echo $ANDROID_HOME` (macOS/Linux) or `echo %ANDROID_HOME%` (Windows)
- Should print your SDK path, not blank

Step 5 — Install Claude Code

Claude Code is a command-line tool by Anthropic. It's what actually writes and builds your project.

1 Install Node.js first (Claude Code requires it)

- Go to: <https://nodejs.org>
- Download the LTS version (left button)
- Install it — click through the defaults
- Verify: open terminal and run: `node --version`
- Should print something like: `v20.x.x`

2 Install Claude Code

- Open terminal
- Run: `npm install -g @anthropic-ai/claude-code`
- Wait for it to finish
- Verify: run `claude --version`

```
$ npm install -g @anthropic-ai/claude-code
$ claude --version
1.x.x
```

3 Log in to Claude Code with your Anthropic account

- Run: `claude`
- It will open a browser window and ask you to log in
- Log in with your Anthropic / Claude account
- Once logged in, come back to terminal — you're ready

Step 6 — Accept Android SDK Licenses

The Android build system requires you to accept Google's licenses before it will compile anything. This is a one-time step.

Open your terminal and run:

```
$ $ANDROID_HOME/cmdline-tools/latest/bin/sdkmanager --licenses

# If that path doesn't work, try:
$ $ANDROID_HOME/tools/bin/sdkmanager --licenses
```

It will show several license agreements one by one. Type `y` and press Enter for each one until it says 'All SDK package licenses accepted'.

Step 7 — Create Your Project Folder & Add the README

1 Create a folder for the project anywhere on your computer

- Example: create a folder called `2048_project` on your Desktop
- Or anywhere you like — Downloads, Documents, etc.

2 Copy the README file into that folder

- Take the file: `README_2048_APK.md` (the one Claude generated for you)
- Paste it into your `2048_project` folder
- That's it — this file tells Claude Code exactly what to build

3 Open terminal inside that folder

- Windows: open the folder in File Explorer, click address bar, type `cmd`, Enter
- macOS: right-click the folder → New Terminal at Folder
- Linux: right-click → Open Terminal Here

Step 8 — Run Claude Code & Let It Build

You're in your terminal, inside the `2048_project` folder. Now just run Claude Code:

```
$ claude

# Once Claude Code starts, type this message:
Read README_2048_APK.md and build the entire project exactly as described.
```

Claude Code will:

- ✓ Check your Java and Android SDK setup
- ✓ Create the Android Studio project
- ✓ Write every Kotlin file (game logic, UI, stats, PDF generator)
- ✓ Set up dependencies and Gradle build files
- ✓ Compile and build the release APK
- ✓ Tell you the exact path to your finished .apk file

This process takes 5–15 minutes. Do not close the terminal.

Step 9 — Get the APK onto Your Phone

1 Find the APK file

- After Claude Code finishes it will print the path
- It will be at: `2048_project/app/build/outputs/apk/release/app-release.apk`

2 Transfer to your Android phone (any method works)

- USB cable: plug in phone, copy the .apk file to phone storage
- WhatsApp: send the .apk to yourself in WhatsApp, download on phone
- Google Drive: upload from PC, download on phone
- Bluetooth: share via Bluetooth from PC to phone

3 Allow installing from unknown sources

- On phone: go to Settings → Apps → Special app access
 - Tap Install unknown apps
 - Allow for whichever app you use to open the APK
- (e.g. Files, Downloads, WhatsApp — whichever you used to transfer it)

4 Install and play!

- Tap the .apk file on your phone
- Tap Install
- Open '2048 Dark Edition' from your app drawer
- Share with friends by sending them the same .apk file

Quick Checklist — Before Running Claude Code

- ☐ Android Studio installed and Setup Wizard completed
- ☐ SDK: Android 14 (API 34) installed in SDK Manager
- ☐ SDK: Android 5.0 (API 21) installed in SDK Manager
- ☐ SDK Build-Tools 34.x installed in SDK Manager
- ☐ `java -version` shows `openjdk 17.x` in terminal

- ☐ ANDROID_HOME environment variable set and verified
- ☐ Node.js installed (node --version works)
- ☐ Claude Code installed (claude --version works)
- ☐ Logged in to Claude Code (ran claude once and authenticated)
- ☐ README_2048_APK.md is inside your project folder
- ☐ Terminal is open inside the project folder

Common Issues & Fixes

■ 'java' is not recognized / command not found

Android Studio bundles JDK 17. Add its jbr/bin/ path to your system PATH. See Step 3 for the exact folder location per OS.

■ 'ANDROID_HOME' is blank / not set

You need to set the environment variable AND open a new terminal window after setting it. Old terminal windows don't pick up new variables. See Step 4.

■ SDK licenses not accepted — build fails

Run: sdkmanager --licenses from your Android SDK's cmdline-tools folder and type y for each one. See Step 6.

■ Claude Code says 'command not found'

Make sure Node.js is installed first, then re-run: npm install -g @anthropic-ai/claude-code and open a new terminal window.

■ Build fails with 'SDK location not found'

ANDROID_HOME is not set correctly. Double-check the path has no trailing slash and matches exactly what Android Studio shows in SDK Manager.

■ Phone says 'App not installed'

You need to enable 'Install unknown apps' for the app you used to open the APK. See Step 9.